The $R_{max}(E_L)$ dependencies in Fig. 5(c) reflect the three scenarios described in section 2.2. In fs breakdown, R_{max} increases continuously with E_L from 200 nm to large bubble sizes, initially with steep and later with smaller slope (scenario 1). By contrast, IR ns pulses generate luminescent plasmas and large bubbles with 50 to 200 µm radius already at threshold (scenario 2). In the third scenario, applicable to UV and VIS ns pulses with smooth temporal shape, the $R_{max}(E_L)$ curves feature an abrupt increase in bubble size that correlates with the onset of bright plasma luminescence. We use the terms "small-bubble regime" and "BPL regime" to distinguish the regions before and after the step. At the bubble threshold, nano-bubbles with a radius as small as 200 nm can be produced, just like in fs breakdown. In the small-bubble regime, the $R_{max}(E_L)$ curve increases more slowly than for fs breakdown close to threshold but in the BPL regime it exhibits the same trend as in the other scenarios. Figure 6 shows the same pattern of stepwise tunability in UV ns breakdown for a range of numerical apertures $0.3 < NA < 0.8$. The sharpness of the BPL threshold decreases slightly with decreasing NA from $S = 96.8$ at $NA = 0.8$ to $S = 43.2$ at $NA = 0.3$. At the same time, the relative jump in bubble radius becomes smaller; the ratio $R_{max>BPL}/R_{max<BPL}$ drops from a value of 10 at NA = 0.8 to a value of 4 at $NA = 0.3$. Here $R_{max<BPL}$ and $R_{max>BPL}$ denote the radii of the expanded bubble just below and above the BPL threshold.

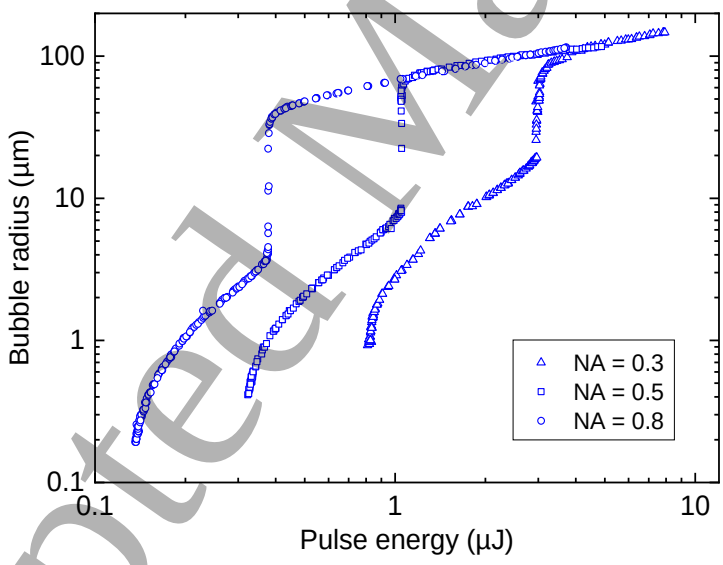

Figure 6 $R_{max}(E_L)$ curves for slm UV laser pulses (355 nm) of 0.56 ns duration at different NAs.

The width of the small-bubble regime for UV and VIS ns pulses increases with laser pulse duration [Fig. 5(c)]. For 560-ps and 1-ns pulses, it comprises an energy range of $\approx 3\times E_{thB}$, while for 8-ns pulses the range is $\approx 20\times E_{thB}$. The nanoregime is broader for ns pulses than for fs breakdown. We use the normalized energy range with bubble sizes between 200 nm and 3 µm, $(E_{L3\mu m}-E_{L200nm})/E_{L200nm}$, as a measure for the width of the nanoregime. For UV wavelengths, this ratio has a value of 0.37 for 280-fs pulses and increases by a factor of four to 1.46 for 560-ps pulses. For 6.8-ns pulses at 355 nm and 8.8-

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ns pulses at 532 nm, the small-bubble regime is even larger but the minimum bubble size is not quite as small as with fs and sub-ns pulses.

Although the scaling laws for plasma-mediated energy deposition represented by the $R_{max}(E_L)$ curves are complex in the vicinity of the bubble threshold, a similar energy scaling is observed for all pulse durations and wavelengths at pulse energies well above threshold. The straight line with slope 1/3 in the log-log plot represents a dependence $R_{max} \propto (E_L)^{1/3}$, which according to Eq. (8) corresponds to $E_B \propto E_L$.

4.3. Plasma luminescence and scattering

Figure 7 presents time-integrated photographs of the shape and inner structure of plasma luminescence with 0.6 µm spatial resolution. Luminescence of fs laser induced plasmas could not be seen with the naked eye or after single photographic exposures but only after integration over 70 breakdown events. Weak luminescence appeared at about $3\times E_{thB}$ and gradually increased with E_L (Fig. 7a). Luminescence from fs plasmas stayed confined to the laser cone angle, whereas the luminescence of BPL ns plasmas extended for all wavelengths up to 20 µm beyond the laser focus and outside the cone angle of the laser beam (Figs. 7 (b) – (e)).

Luminescence from UV ns breakdown is shown in Fig 7b. In the small-bubble regime [left column of (b)], no luminescence is observed and the plasma becomes visible only through scattering of the incident laser light. It appears green because the camera chip is not sensitive at UV wavelengths but can detect 2nd harmonic radiation that leaked through the wavelength separator after third-harmonic generation. The abrupt onset of bright plasma luminescence at E_L = 45 µJ (right column of Fig 7b) coincides with the sharp increase of the bubble radius seen in Figs. 5 and 6. At the BPL threshold, thermal ionization becomes sufficiently strong to overcome the inhibiting action of recombination, as described in section II.B. The sharpness of this threshold is similar to that for bubble formation (Table 2).

The photographs of plasmas produced by 532-nm ns pulses show scattered pump laser light emanating from a region within the cone angle of the laser beam, which is surrounded by a luminescent halo [Figs. 7 (c), (e), and (f)]. The scattering region defines the region of primary energy deposition by the incident laser light. It consists of branched strings with ≈ 1 µm diameter that become brighter in upstream direction. Oversaturation from scattered pump laser light is avoided by blocking filters. The dielectric filter used in Fig. 7(e) and (f) had 0.03 % transmission at 532 nm, about 1/10 of the transmission of the colour glass filter used in Fig. 7(c). This revealed the string-like substructure of the primary-energy-deposition region, which cannot be resolved in Fig. 7(c).

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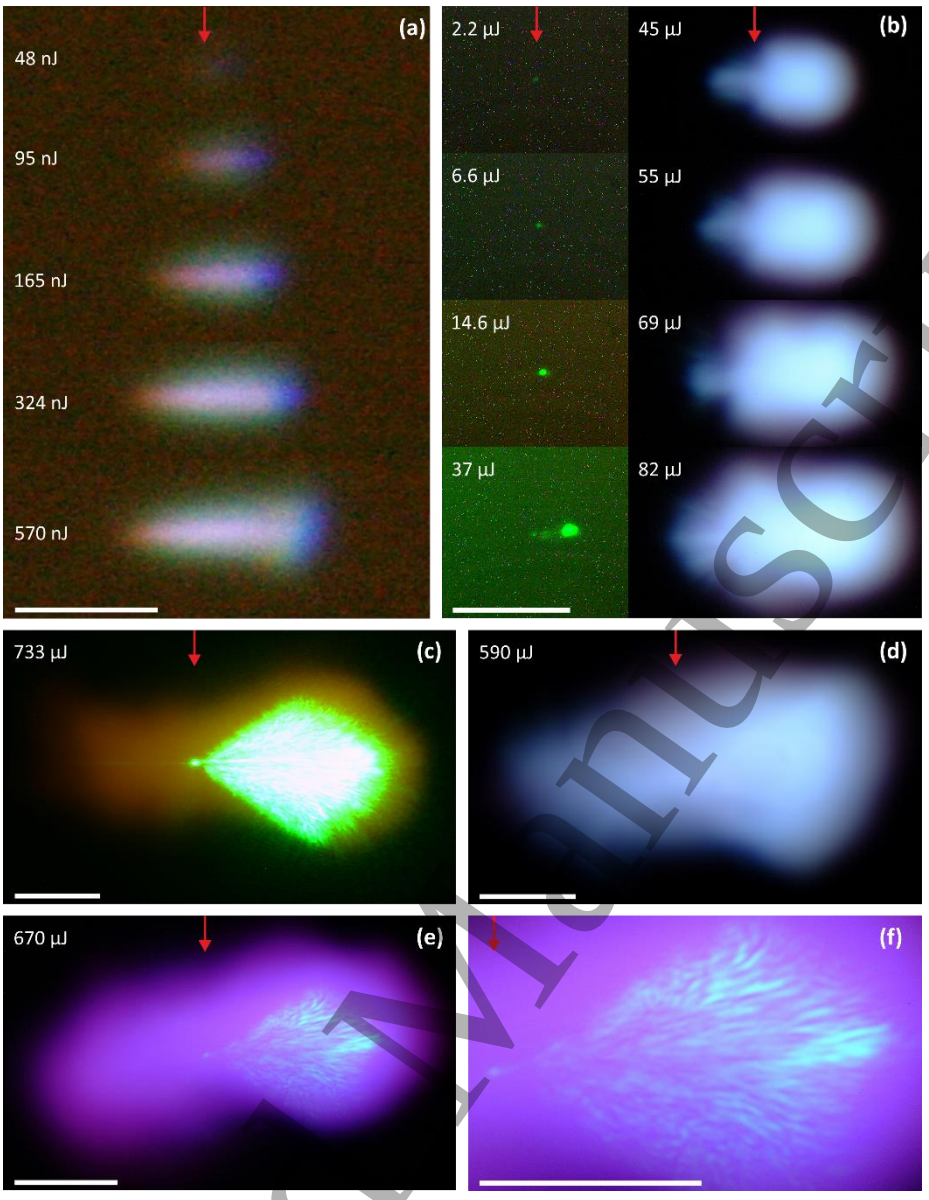

Figure 7 Side-view photographs of plasmas in water produced at NA = 0.8 by temporally smooth laser pulses of different duration and wavelength: 350-fs, 1040 nm in (a), 6.8-ns 355 nm in (b), 8.8 ns, 532nm in (c), (e) and (f), and 11.2 ns, 1064 nm in (d). The corresponding pulse energies are given in each frame. Laser light is incident from the right, and the position of the beam waist is marked by an arrow. Luminescence is integrated over 70 pulses in (a) and over 100 pulses in the left column of (b), both at ISO 3200. All other photographs are from single laser pulses. In (b), the laser cone angle is depicted in the picture taken at E_L = 82 µJ. In (c), (e), and (f), the camera detected both plasma luminescence and scattered laser light from the 532-nm ns pulse, whereby (f) shows an enlarged view of the central region of (e). Scattered laser radiation was in (c) partially blocked by a long-pass colour glass filter, and in (e), (f) by a dielectric notch filter. The scattered light demarcates the region of primary energy deposition within the laser cone angle. Redistribution of the plasma energy then causes the luminescence emanating from a larger volume that partly lies outside the region reached by the laser beam. This phenomenon is visible also in (b) and (d). In (d) the plasma core can be identified by its stronger luminescence. Length of scale: 5 µm in (a), and 20 µm in (b) - (f). In (e) and (f) one can see bright strings within the plasma core that have diameters in the order of 1 µm. These strings are indicative for strong scattering of the incoming laser light and, thus, demarcate zones of particularly intense primary energy deposition. The strings are only visible at 532 nm pump laser wavelength because the CMOS camera chip does not detect the light at 355 nm and 1064 nm.

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